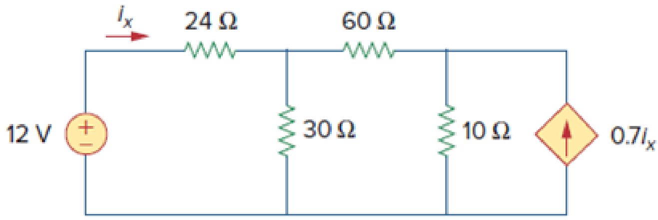
Problem

Use source transformation on the circuit shown in Fig 4.98 to find *ix*.

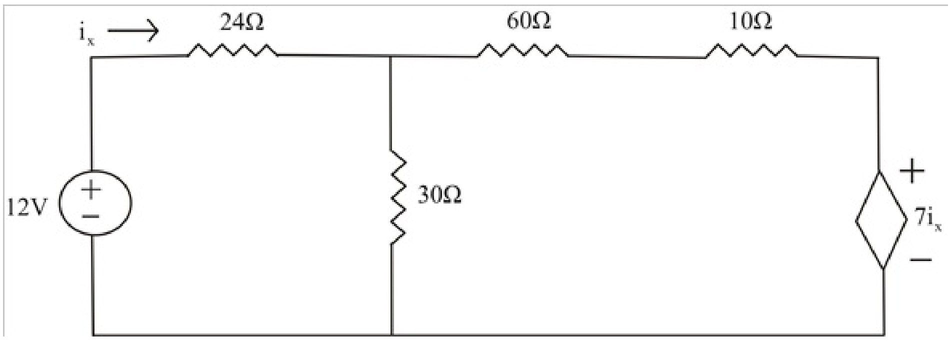
Figure 4.98:



Step-by-step solution

Step 1 of 5

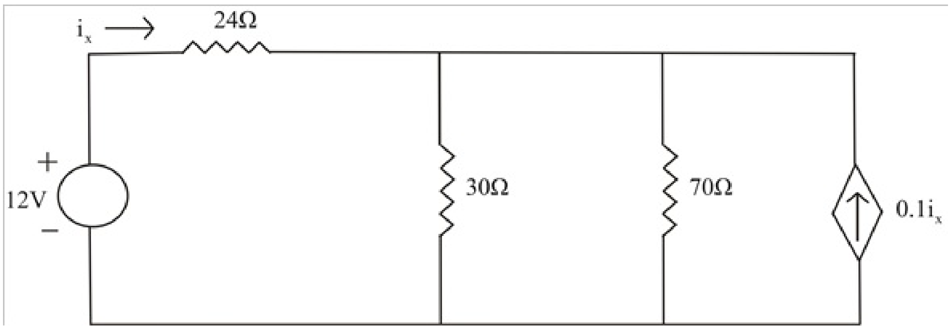
Transform the given current source as shown below.



Step 2 of 5

Combine the 60Ω with the 10Ω and transform the dependent

Source as shown below.



Step 3 of 5

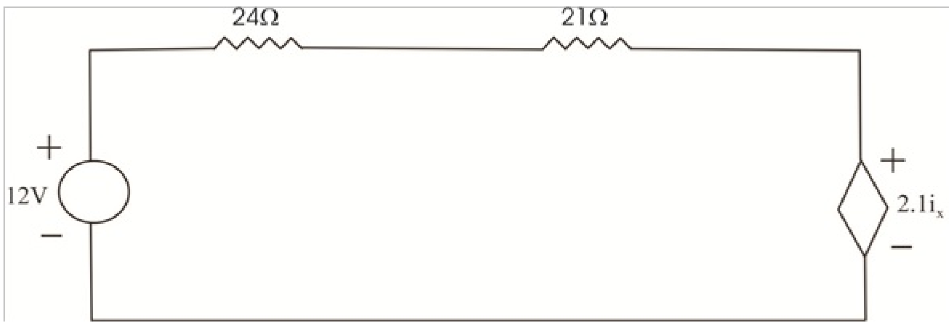
Combining 30Ω and 70Ω gives,

$$30 \| 70 = 70 \times \frac{30}{100}$$

$$= 21 \text{ Ohms}$$

Transform the dependent current source as shown below.

Step 4 of 5



Step 5 of 5

Applying the KVL to the loop gives.

$$45i_x - 12 + 2.1i_x = 0$$

$$i_x = \frac{12}{47.1}$$

$$= 254.8\,\text{mA}$$